Topic 40 — Desertification (मरुस्थलीकरण)

Covers syllabus (click to expand)

Desertification | Land Degradation (भू-क्षरण) | Drought (सूखा) | Soil Erosion

Sources baked in: NCERT; ISRO Desertification Atlas 2016; UNCCD; IMD; DPAP; UPPCS PYQs **Weight:** ★★ Medium — ASDM definition; sheet→ravine; UNCCD; LDN 2030; DPAP **Last verified:** August 2026

Current Affairs (this topic)

- Twin of Desertification & Land Degradation — UNCCD / LDN CA refresh.
- Keep desertification ≠ drought-only wording clear.

40.1 Desertification

Identity: ASDM drylands | Productivity loss | Overgrazing | UNCCD 1994

- **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation and human activities.
- It is **not** only the physical expansion of sand deserts like the Thar. Any qualifying dryland that loses fertility and vegetation can desertify.
- **Overgrazing** is the leading human cause in drylands; deforestation and unsustainable cropping follow.
- Improper irrigation can cause **salinization** and **waterlogging**.
- Drought and climate stress kill vegetation and speed erosion — a feedback loop.
- **Wind erosion** dominates true arid zones such as the **Thar** (थार).
- India's **ISRO Desertification Atlas (2016)** mapped about **96.4 M ha (~29.3%)** under desertification/land degradation; **Rajasthan** is the worst-affected state.
- **UNCCD** adopted **17 June 1994** (Paris); force **1996**.
- **17 June** = World Desertification and Drought Day.
- **SDG 15.3** targets **Land Degradation Neutrality (LDN) by 2030**.

PYQ — UPPCS Prelims 2019, Q42

India aims to achieve land degradation neutrality by the year

Options: A. 2025 B. 2030 C. 2035 D. 2040

Show answer

Ans: B

> **Logic:** Bundelkhand (dry sub-humid) can desertify — not only sand deserts.

40.2 Land Degradation

Identity: Broader than desertification | Processes | Wasteland | Watershed fix

- **Land degradation** is any lasting decline in the land's productive capacity.
- **Desertification** is the ASDM subset; erosion, salinity, waterlogging, mining, and pollution are processes under the broader term.
- Continued degradation produces **wasteland** unsuitable for cultivation.
- Consequences: food insecurity, rural poverty, migration, reservoir silting, and higher flood risk on silted beds.
- Solutions: watershed management (वाटरशेड (जलग्रहण) प्रबंधन), afforestation, controlled grazing, soil-water conservation, and LDN restoration projects.
- **Afforestation reduces** degradation — it is not a cause.

Process	Meaning
Water/wind erosion	Topsoil loss
Salinization	Excess irrigation salts
Waterlogging	Poor drainage
Mining / pollution	Direct land damage

> **Logic:** UNCCD ≠ UNFCCC. LDN year = **2030**.

40.3 Drought

Identity: Water deficit types | IMD LPA | DPAP 1973 | Bundelkhand link

- **Drought** is a prolonged water deficit relative (सापेक्ष) to normal supply — not the same as desertification, though the two reinforce each other.
- **Meteorological drought:** rainfall well below normal (IMD often uses **< 75% of Long Period Average** as a severe marker).
- **Hydrological drought (जल-वैज्ञानिक सूखा):** rivers, reservoirs, and groundwater fall.
- **Agricultural drought (कृषि सूखा):** soil moisture fails crops even if some rain fell.
- Socio-economic drought (सामाजिक-आर्थिक सूखा) follows when livelihoods and prices collapse.
- Drought kills vegetation → bare soil → erosion → desertification risk rises.

- **Drought Prone Areas Programme (DPAP)** launched **1973–74** for dryland watershed-type works.
- Related lineage: Desert Development Programme (DDP), later watershed/IWMP-type convergence.
- UP hotspot: **Bundelkhand** seven districts — Jhansi (झांसी), Lalitpur, Jalaun, Hamirpur, Mahoba, Banda, Chitrakoot (चित्रकूट).

PYQ — UPPCS Prelims 2025, Q134

Arrange chronologically: 1. Food for Work 2. Community Development Programme 3. TRYSEM 4. Drought Prone Areas Programme

Options: A. 4, 2, 3, 1 B. 2, 4, 1, 3 C. 2, 4, 3, 1 D. 4, 2, 1, 3

Show answer

Ans: B — CDP (सीडीपी) **1952** → DPAP **1973** → Food for Work **1977** → TRYSEM **1979**.

> **Logic:** Drought ≠ famine automatically; drought ≠ permanent desert.

40.4 Soil Erosion

Identity: Normal vs accelerated | Sheet→ravine | Wind | Conservation

- **Normal (geological) erosion** is slow and roughly balances soil formation.
- **Accelerated erosion** is human-driven and faster than formation — the focus.
- **Water erosion ladder:sheet → rill → gully → ravine**.
- **Chambal ravines** (चंबल) (MP–UP–Rajasthan, including Etawah (इटावा)–Agra belt) are the classic Indian ravine badlands.
- **Wind erosion:deflation** (lifting fine particles) and **abrasion** (sandblasting) — strong in the Thar.
- Conservation: contour bunding, terracing, shelter belts, check dams, gully plugging, cover crops, and afforestation.
- Healthy soil biology (e.g. earthworms) supports fertility and structure.

PYQ — UPPCS Prelims 2018, Q101

Thar / Rajasthan desert expanse is mainly of:

Options: A. Pliocene B. Paleocene C. Pleistocene and recent deposits D. Oligocene

Show answer

Ans: C

Type	Feature
Sheet	Uniform thin topsoil loss
Rill	Small channels

Type	Feature
Gully	Deeper channels
Ravine	Deep badlands (Chambal)
Wind	Deflation + abrasion

> **Logic:** Afforestation and shelter belts **prevent** erosion — they are not causes.
